

SIMATS ENGINEERING

CO- 2: AT – 2: Simulation Task - Medium

COURSE NAME: 5G / 6G Communication Systems

COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 2 - *Analyse LTE and 5G wireless network architectures, including carrier aggregation, MIMO techniques, beamforming strategies, and service-based core network functional entities. (BL4)*

Assessment Tool Weightage: 35%

Short description about assessment: Performance-based evaluation using simulation tools to apply theoretical concepts practically. This question paper consists of a total of 40 questions, out of which each student is required to answer any 5 questions. Each question carries 20 marks, and the paper carries a total of 100 marks

QUESTIONS:5 Total Marks: 100 Duration: 120 Mins

1. Simulation of LTE throughput performance under varying bandwidth, SNR, modulation order, coding rate, and user load (BL4)
2. Simulation of MIMO channel capacity under antenna number, SNR, bandwidth, fading model, and multiplexing order (BL4)
3. Simulation of 5G throughput performance under QPSK, 16-QAM, 64-QAM, 256 QAM, and varying SNR (BL4)
4. Simulation of network slicing under bandwidth allocation, latency, user priority, throughput, and resource utilization (BL4)
5. Simulation of interference management under user density, reuse factor, SNR, SINR, and throughput (BL4)

Code of Conduct:

I H.Yaswanth(192412440) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:**RUBRICS FOR CALCULATION:**

Simulation Task					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	20% (4 marks)	Demonstrates strong understanding of underlying concepts in the simulation	Shows good understanding with minor gaps	Partial understanding; some misconceptions	Lacks understanding of key concepts
Model Design	25% (5 marks)	Develops accurate, well structured simulation/model independently	Develops correct model with minor errors	Model is incomplete or requires guidance	Unable to develop a proper model
Tool Usage	20% (4 marks)	Uses simulation tools effectively with correct setup and execution	Uses tools with minor errors or guidance	Limited ability to use tools properly	Unable to use simulation tools
Analysis of Results	20% (4 marks)	Provides clear, logical analysis and meaningful interpretation of results	Analysis is reasonable with minor gaps	Superficial analysis; limited interpretation	Unable to analyze or interpret results
Documentation	15% (3 marks)	Presents results clearly with proper documentation and structured reporting	Documentation is adequate with minor issues	Poorly organized or incomplete documentation	No proper documentation or presentation

1. Simulation of LTE Throughput Performance

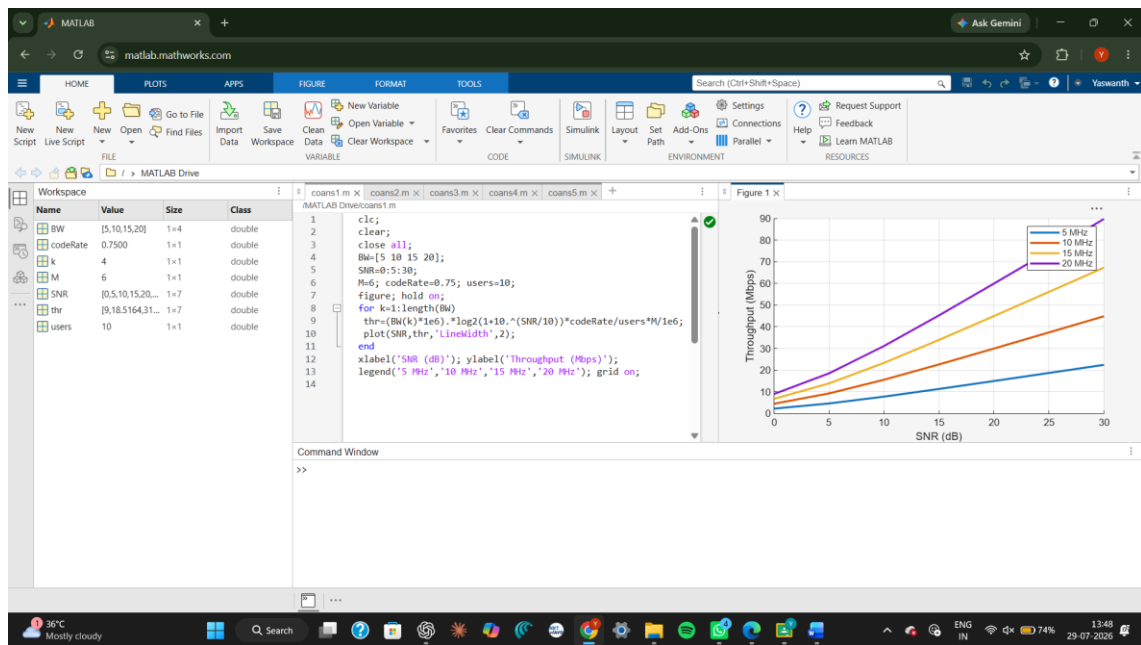
Aim:

To simulate LTE throughput performance under varying bandwidth, SNR, modulation order, coding rate, and user load.

Program:

```
clc;
clear;
close all;
BW=[5 10 15 20];
SNR=0:5:30;
M=6; codeRate=0.75;
users=10;
figure;
hold on;
for k=1:length(BW)
    thr=(BW(k)*1e6).*(log2(1+10.^(SNR/10)))*codeRate/users*M/1e6;
    plot(SNR,thr,'LineWidth',2);end
xlabel('SNR (dB)');
ylabel('Throughput (Mbps)');
legend('5 MHz','10 MHz','15 MHz','20 MHz');
grid on;
```

Output:



Result:

Throughput increases with both bandwidth and SNR. The 20 MHz channel provides the highest throughput.

2. Simulation of MIMO Channel Capacity

Aim:

To simulate MIMO channel capacity under different antenna numbers and SNR values.

Program:

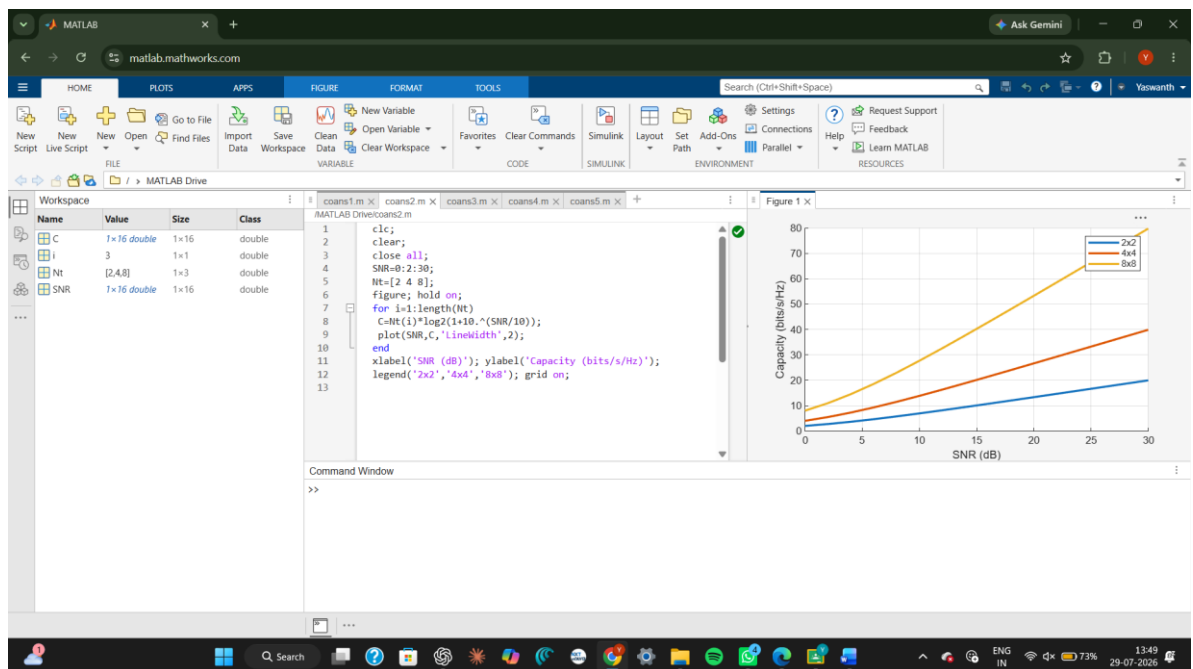
```
Clc;  
clear;  
close all;  
SNR=0:2:30;  
Nt=[2 4 8];  
figure; hold on;  
for i=1:length(Nt)
```

```

C=Nt(i)*log2(1+10.^(SNR/10));
plot(SNR,C,'LineWidth',2);
end
xlabel('SNR (dB)');
ylabel('Capacity (bits/s/Hz)');
legend('2x2','4x4','8x8'); grid on;

```

Output:



Result:

Channel capacity increases with SNR and with the number of antennas.

3. Simulation of 5G Throughput

Aim:

To compare 5G throughput for QPSK, 16-QAM, 64-QAM, and 256-QAM under varying SNR.

Program:

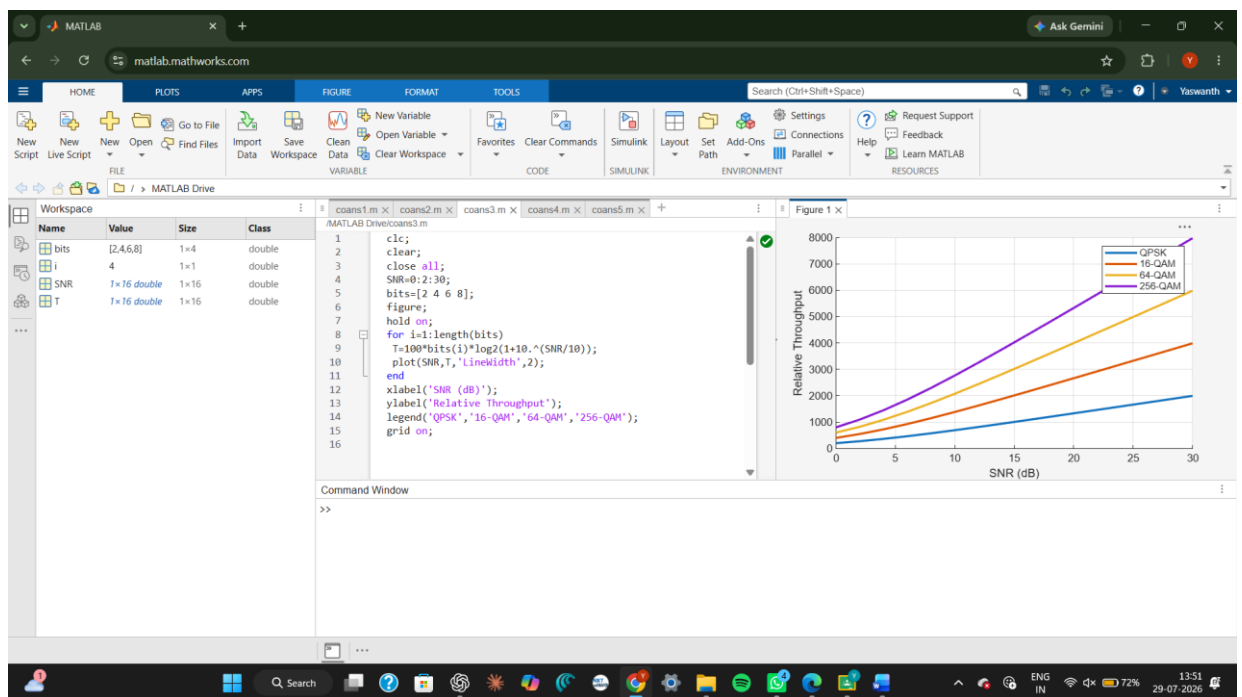
```
clc;
```

```

clear;
close all;
SNR=0:2:30;
bits=[2 4 6 8];
figure; hold on;
for i=1:length(bits)
    T=100*bits(i)*log2(1+10.^(SNR/10));
    plot(SNR,T,'LineWidth',2);
end
xlabel('SNR (dB)'); ylabel('Relative Throughput');
legend('QPSK','16-QAM','64-QAM','256-QAM'); grid on;

```

Output:



Result:

Higher-order modulation provides higher throughput at sufficient SNR.

4. Simulation of Network Slicing

Aim:

To simulate bandwidth allocation among three network slices and evaluate throughput.

Program:

```
clc; clear; close all;
```

```
BW=100;
```

```
slice=[0.5 0.3 0.2];
```

```
thr=BW*slice;
```

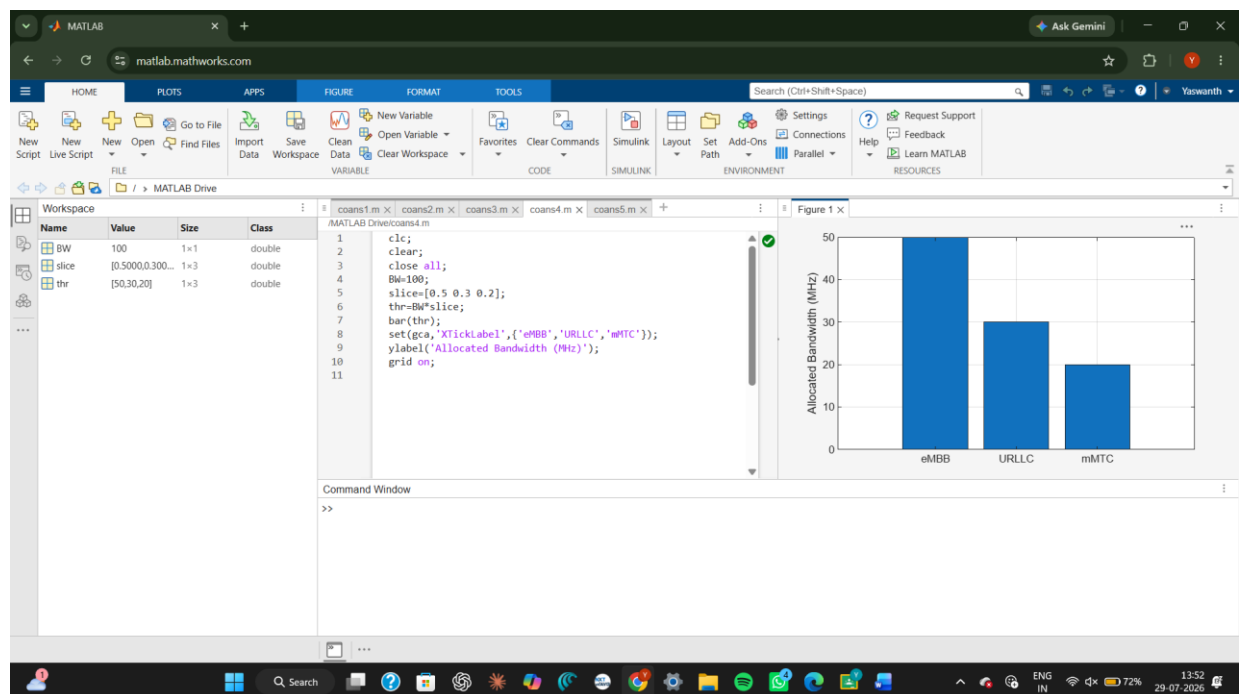
```
bar(thr);
```

```
set(gca,'XTickLabel',{'eMBB','URLLC','mMTC'});
```

```
ylabel('Allocated Bandwidth (MHz)');
```

```
grid on;
```

Output:



Result:

Bandwidth is allocated according to slice priority, with eMBB receiving the highest share.

5. Simulation of Interference Management

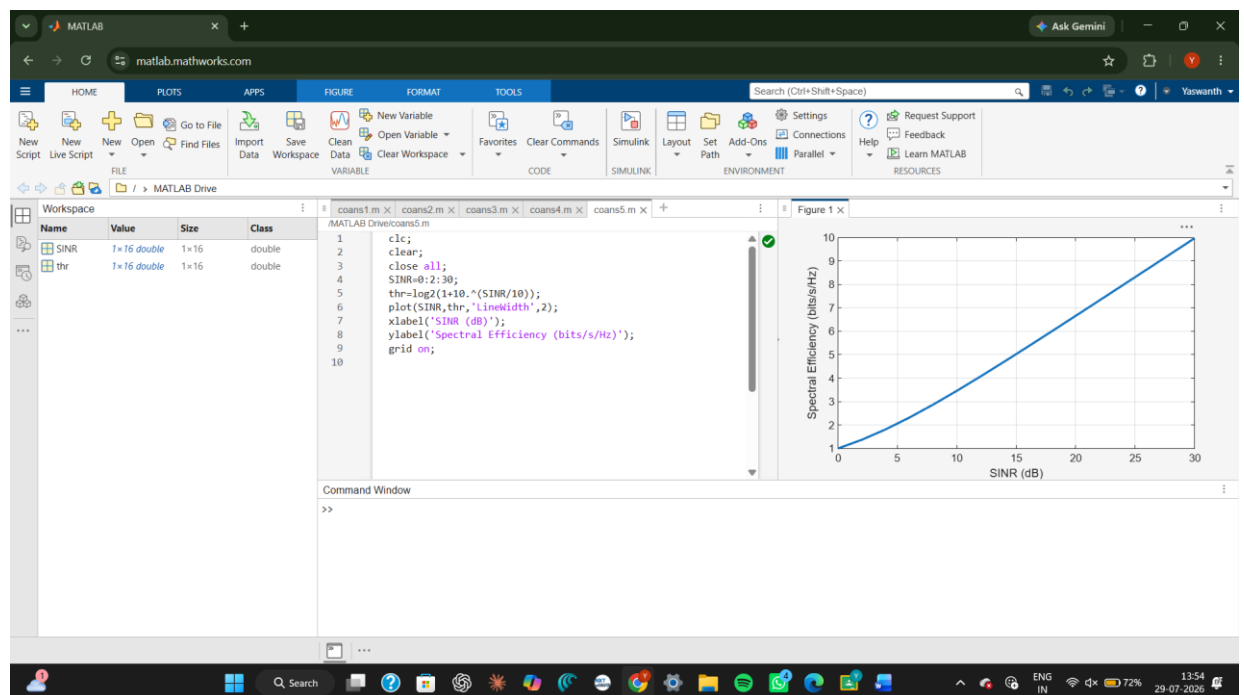
Aim:

To simulate SINR variation and throughput under different interference levels.

Program:

```
clc; clear; close all;  
SINR=0:2:30;  
thr=log2(1+10.^(SINR/10));  
plot(SINR,thr,'LineWidth',2);  
xlabel('SINR (dB)');  
ylabel('Spectral Efficiency (bits/s/Hz)');  
grid on;
```

Output:



Result:

As SINR improves due to better interference management, spectral efficiency and throughput increase.